

Phishing Attack Simulation Using Gophish Framework

What is Gophish ?

Gophish is an **open-source phishing framework** used by security professionals to **simulate real-world phishing attacks** in order to:

- Test employee awareness
- Identify weak points in email security
- Train users

Background of Scenario

*An organization wants to test the **phishing awareness of its employees** after noticing an increase in phishing emails impersonating IT support teams.*

*The security team decides to conduct a **controlled phishing simulation** using the **Gophish framework** to evaluate how employees respond to suspicious emails.*

```
Start Campaign
  ↓
Email Sent
  ↓
User Opens Email?
  ↓
Clicks Link?
  ↓
Credentials Submitted?
  ↓
Data Logged in Gophish
  ↓
Report Generated
  ↓
Awareness Training
  ↓
End
```

After the phishing campaign is completed, Gophish records all user interactions such as email opens, link clicks, and credential submissions.

The security team analyzes the data to identify vulnerable users and departments. Based on the findings, awareness training is conducted, email security controls are improved, and future phishing simulations are planned.

Requirements

Red teamers use specialized phishing simulation tools to carry out campaigns safely and effectively. Common tools include `GoPhish` , `King Phisher` , `PhishMe` , and `LUCY` , which allow teams to craft realistic phishing emails, track clicks, and analyze employee responses without compromising real data.

For example, using `GoPhish` , a red team can send a simulated email with a fake login page, monitor who interacts with it, and generate reports on which employees or departments are most vulnerable, helping the organization strengthen training and technical defenses.

Setup

Link : <https://github.com/gophish/gophish/releases/>



step 1: open link

step 2: Download For Windows

Step 3 : Unzip file

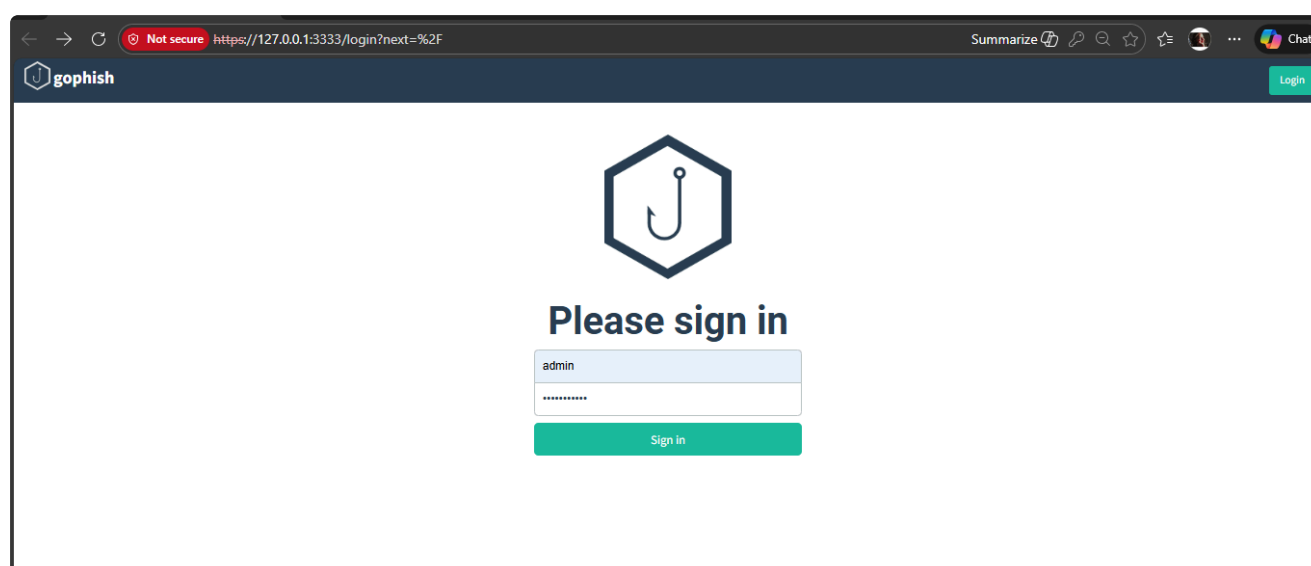
Name	Date modified	Type	Size
db	05-12-2025 17:13	File folder	
static	05-12-2025 17:13	File folder	
templates	05-12-2025 17:13	File folder	
config	05-12-2025 17:13	JSON Source File	1 KB
gophish	08-01-2026 18:55	Data Base File	128 KB
gophish	05-12-2025 17:13	Application	33,780 KB
gophish_admin	05-12-2025 17:14	Security Certificate	1 KB
gophish_admin.key	05-12-2025 17:14	KEY File	1 KB
LICENSE	05-12-2025 17:13	File	2 KB
README	05-12-2025 17:13	MD File	4 KB
VERSION	05-12-2025 17:13	File	1 KB

Step 4 : Run `gophish.exe`

```
time="2026-01-10T14:14:05+05:30" level=warning msg="No contact address has been configured."
time="2026-01-10T14:14:05+05:30" level=warning msg="Please consider adding a contact_address entry in your config.json"
goose: no migrations to run. current version: 20220321133237
time="2026-01-10T14:14:05+05:30" level=info msg="Starting IMAP monitor manager"
time="2026-01-10T14:14:05+05:30" level=info msg="Starting admin server at https://127.0.0.1:3333"
time="2026-01-10T14:14:05+05:30" level=info msg="Starting new IMAP monitor for user admin"
time="2026-01-10T14:14:05+05:30" level=info msg="Starting phishing server at http://0.0.0.0:80"
time="2026-01-10T14:14:05+05:30" level=info msg="Background Worker Started Successfully - Waiting for Campaigns"
```

Step 5 : It will launch the server at `https://127.0.0.1:3333` (Press ctrl and right click by cursor)

Note - Don't Close this terminal



Step 6 : For Credential Go to terminal

```
j="Please login with the username admin and the password 303749d0d5a2ec9d"
j="Starting phishing server at http://0.0.0.0:80"
j="Background Worker Started Successfully - Waiting for Campaigns"
j="Starting IMAP monitor manager"
j="Creating new self-signed certificates for administration interface"
j="Starting new IMAP monitor for user admin"
j="TLS Certificate Generation complete"
```

- **Use this username and password for 1st time login**

Step 7 : After authentication you will be redirected to `set new password` page

Reset Your Password

Password

Confirm Password

Save Password

- *Here we are on our dashboard of gophish*

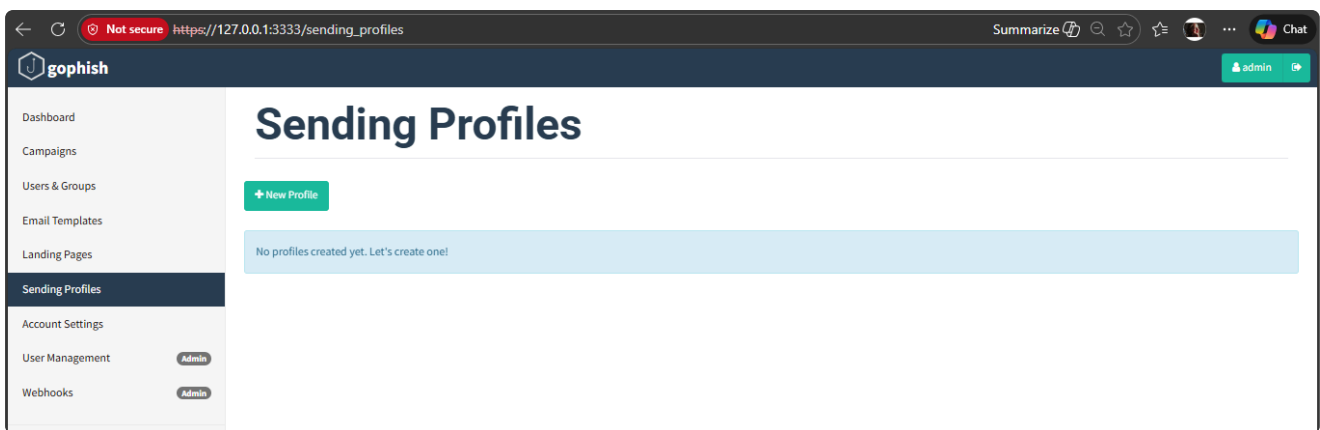
Dashboard

No campaigns created yet. Let's create one!

Launching a phishing Campaign

The phishing campaign was launched using the Gophish dashboard after completing all configurations. Emails were sent to selected users, and their interactions were tracked in real time. The campaign ran for a predefined duration, and the collected results were used for analysis and awareness training.

Sending Profiles : A sending profile in Gophish is used to configure SMTP email settings required to send phishing emails. It includes sender details and mail server configuration. The sending profile must be tested before launching the campaign to ensure proper email delivery.



- **Click on New Profile.**

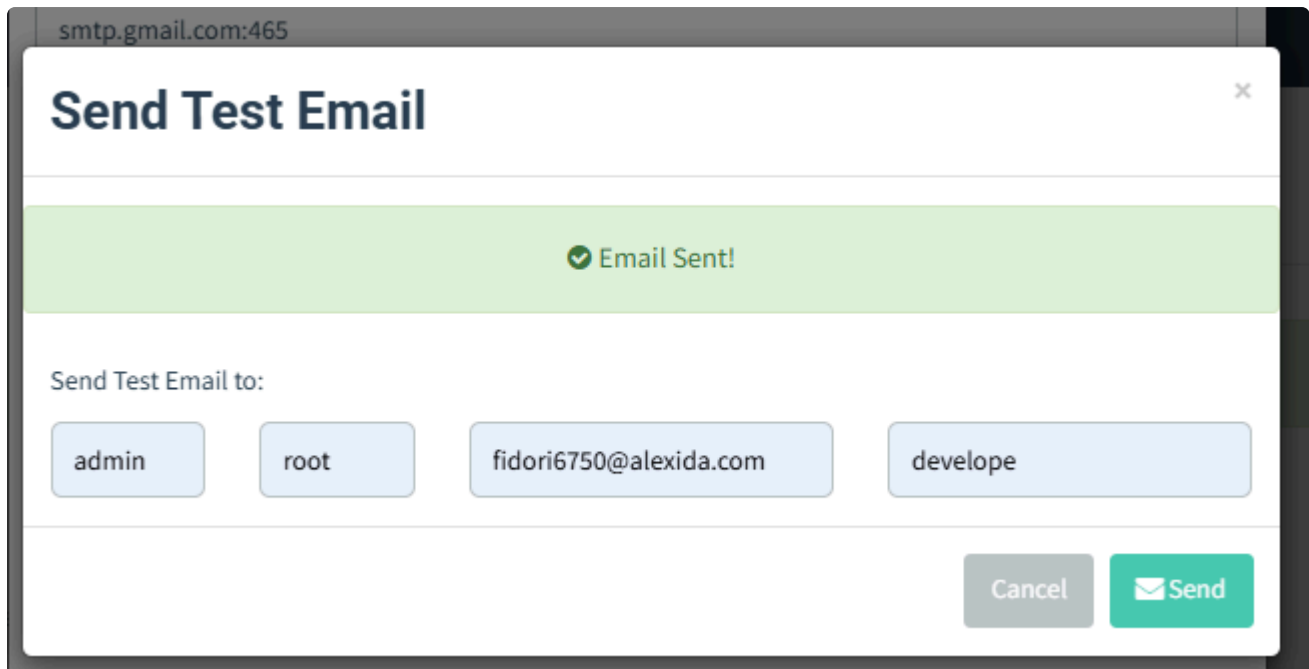
- **From Name** – Display name of the sender
- **From Email** – Sender's email address
- **Host** – SMTP server address
- **Port** – SMTP port number
- **Username & Password** – SMTP authentication
- **TLS Settings** – Secure email transmission

App password is required when using Gmail SMTP

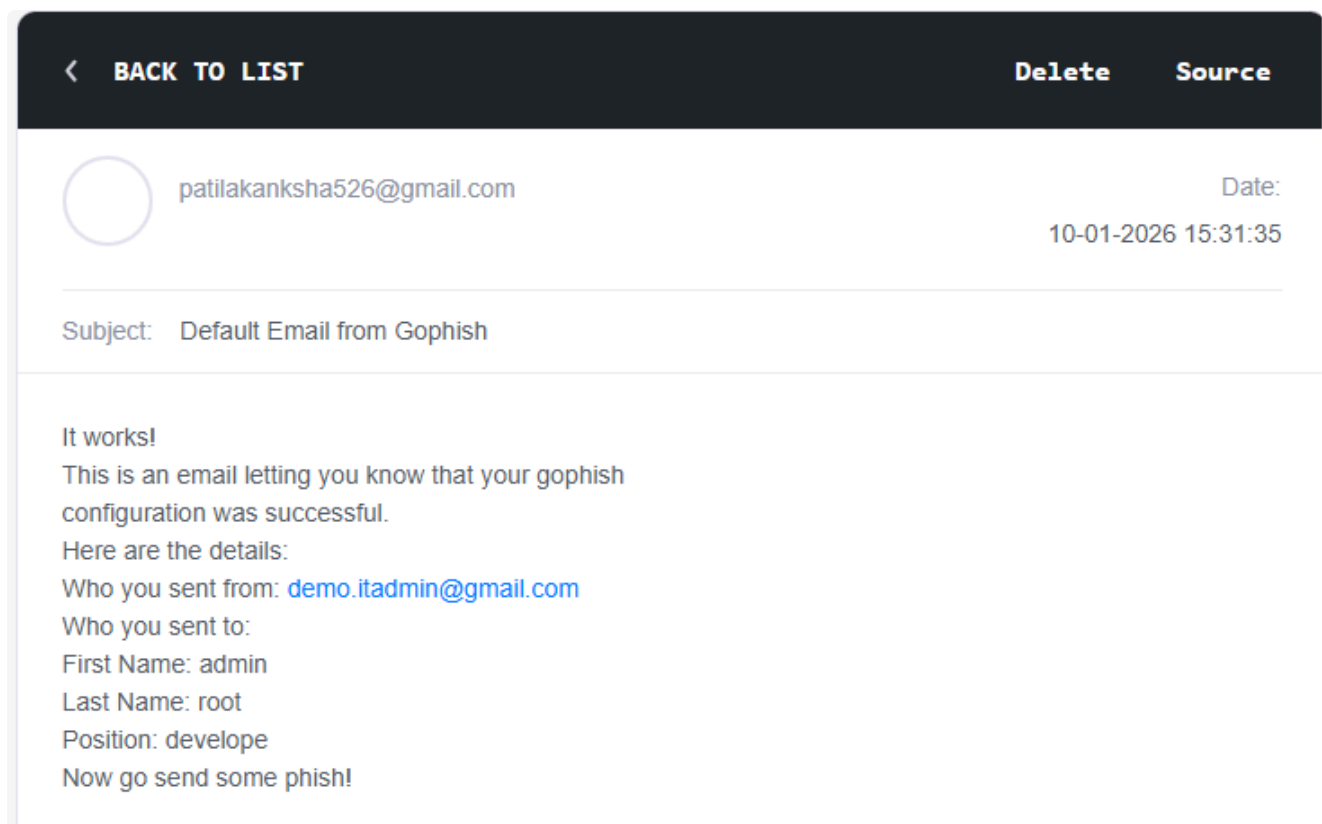
A sending profile defines SMTP settings in Gophish. Gmail SMTP uses an App Password for

authentication. After configuring and testing the profile, it is used to launch the phishing campaign.

- After Setting up every thing try to send test mail to verify everything is working smoothly.



A screenshot of a web application interface showing a 'Send Test Email' dialog box. The dialog has a title bar with 'smtp.gmail.com:465' and a close button. The main content area has a green banner with a checkmark and the text 'Email Sent!'. Below this, the text 'Send Test Email to:' is followed by four input fields containing 'admin', 'root', 'fidori6750@alexida.com', and 'develope'. At the bottom right, there are 'Cancel' and 'Send' buttons.



A screenshot of an email details view in a web application. The top bar is dark with a back arrow, 'BACK TO LIST', and 'Delete' and 'Source' links. The email is from 'patilakanksha526@gmail.com' with a date of '10-01-2026 15:31:35'. The subject is 'Default Email from Gophish'. The body text reads: 'It works! This is an email letting you know that your gophish configuration was successful. Here are the details: Who you sent from: demo.itadmin@gmail.com Who you sent to: First Name: admin Last Name: root Position: develope Now go send some phish!'

If successful then `save profile`

gophish

admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management Admin

Webhooks Admin

Sending Profiles

Profile edited successfully!

+ New Profile

Show entries

Search:

Name	Interface Type	Last Modified Date
IT Admin	SMTP	January 10th 2026, 3:43:46 pm

Showing 1 to 1 of 1 entries

Previous

1

Next

Email Template : The Email Template in Gophish defines the content and structure of the phishing email sent during the campaign. It includes the subject line, email body, embedded phishing link, and tracking options. The template is designed to simulate a realistic organizational email, such as an IT support notification, to observe user behavior in a controlled environment. Tracking is enabled to monitor email opens and link clicks, and the template is tested before campaign launch to ensure correct formatting, link redirection, and proper functionality.

Edit Template

Name:

Google

Import Email

Envelope Sender: 

patilakanksha526@gmail.com

Subject:

John Smith shared a document

Text

HTML





Dear User,

We detected unusual activity on your email account.

[Verify Email Account](#)

If not verified within 24 hours, access may be suspended.

Regards,

IT Support Team

```
<html>
<body style="font-family: Arial;">

<p>Dear User,</p>

<p>We detected unusual activity on your email account.</p>

<p>
<a href="{{.URL}}">Verify Email Account</a>
</p>

<p>If not verified within 24 hours, access may be suspended.</p>

<p>
Regards,<br>
IT Support Team
</p>

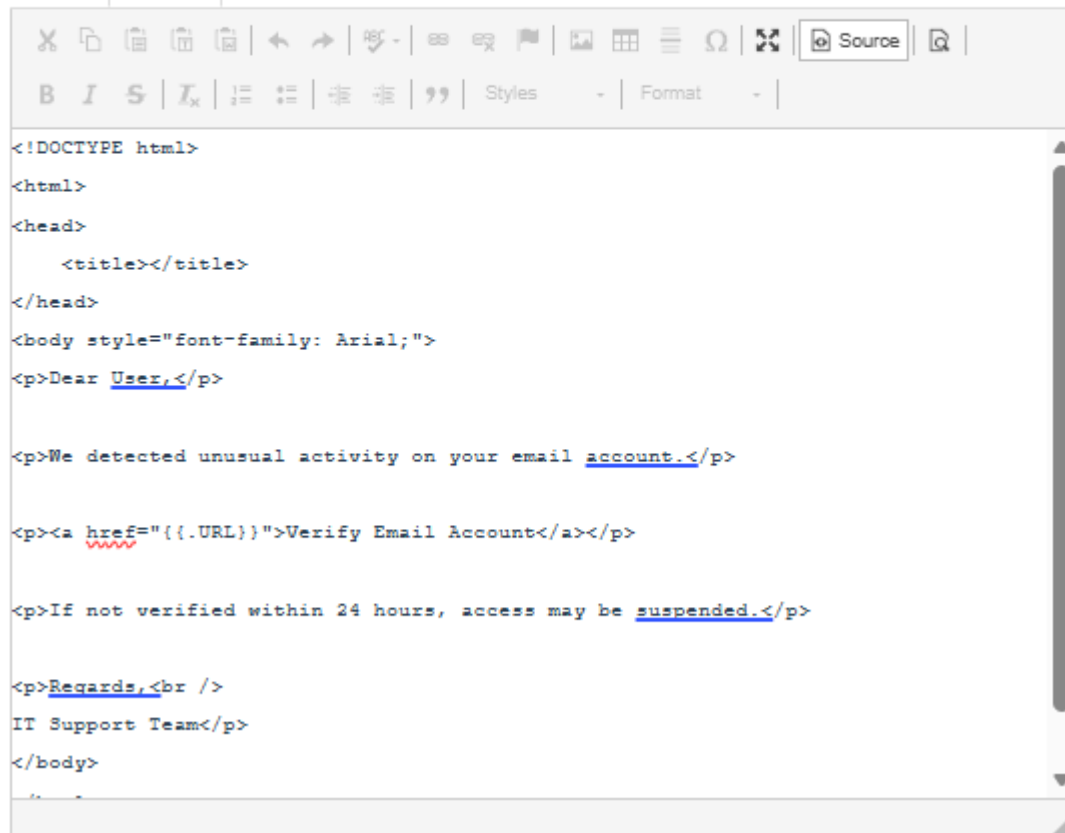
</body>
</html>
```

Subject

John Smith shared a document

Text

HTML

The image shows a screenshot of an email editor interface. At the top, there's a subject line that says "John Smith shared a document". Below it, there are two tabs: "Text" and "HTML". The "HTML" tab is selected. The main area shows the HTML code for the email body. The code includes a DOCTYPE declaration, a head section with a title, and a body section with a font-family style. The body contains several paragraphs of text, including a greeting, a notification about unusual activity, a link to verify the email account, a warning about access suspension, and a sign-off from the IT Support Team. The code is color-coded, with tags in blue and text in black. A vertical scrollbar is visible on the right side of the code editor. Below the code editor, there's a preview area showing the rendered email content, which matches the text in the code editor. The preview shows the text "Dear User," followed by a link "Verify Email Account", then "We detected unusual activity on your email account.", then "If not verified within 24 hours, access may be suspended.", and finally "Regards," followed by "IT Support Team" on a new line.

```
<!DOCTYPE html>
<html>
<head>
  <title></title>
</head>
<body style="font-family: Arial;">
<p>Dear User,</p>

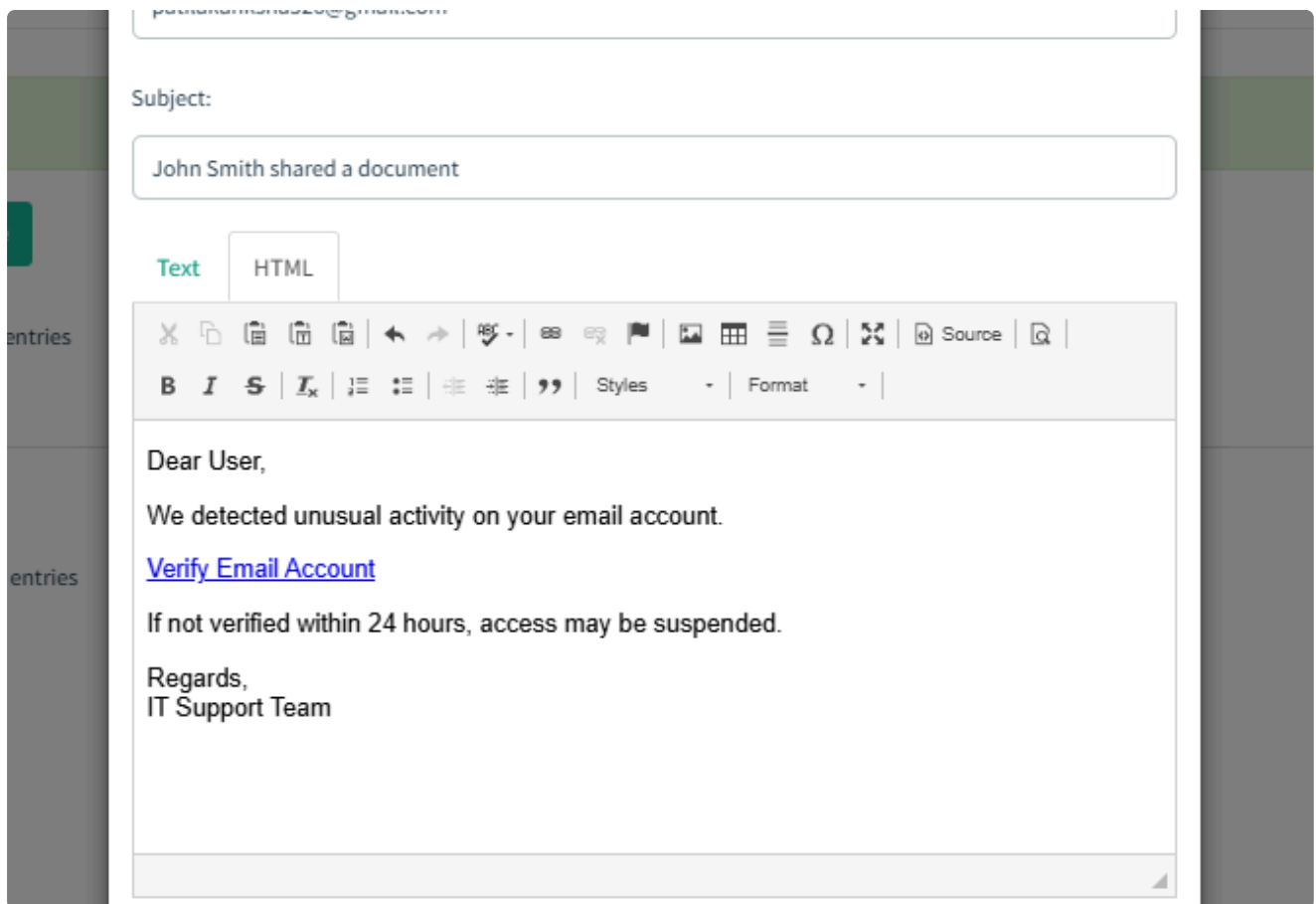
<p>We detected unusual activity on your email account.</p>

<p><a href="{{.URL}}">Verify Email Account</a></p>

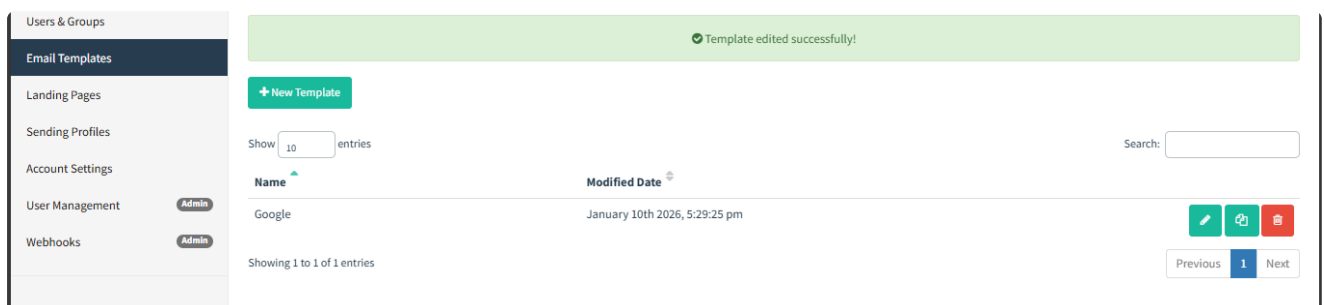
<p>If not verified within 24 hours, access may be suspended.</p>

<p>Regards,<br />
IT Support Team</p>
</body>
```

Click on Source to Render



Save Template



Landing Page : The Landing Page in Gophish is the web page to which users are redirected after clicking the phishing link in the email. It is designed to simulate a legitimate login or verification page in a controlled environment. The landing page is configured to capture user interactions such as page visits and form submissions, while ensuring no real harm is caused. This component helps in analyzing user behavior and measuring the effectiveness of the phishing campaign.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <style>
  body {
    margin: 0px;
  }
  /* Background */
  #background {
```

```
    position: absolute;
    width: 100%;
    height: 100%;
    background: linear-gradient(106.75deg, #FFE7E7 1.62%, rgba(255, 253, 213,
0) 96.07%);
}
```

```
/* Window */
#window {
    position: absolute;
    background: #ffffff;
    width: 450px;
    height: 400px;
    top: 30%;
    left: 40%;
    -webkit-box-shadow: 5px 10px 6px -6px #777;
    -moz-box-shadow: 5px 10px 6px -6px #777;
    box-shadow: 5px 10px 6px -6px #777;
}
```

```
/* Microsoft 365 Logo */
#microsoft-logo {
    position: absolute;
    width: 190px;
    height: 51px;
    margin: 15px 0 0 15px;
}
```

```
/*Confirm Details*/
#confirm-details {
    position: absolute;
    width: 294px;
    height: 26px;
    font-family: Calibri;
    font-style: normal;
    font-weight: normal;
    font-size: 24px;
    line-height: 28px;
    color: #000000;
    margin: 100px 0 0 28px;
}
```

```
#email-container {
    margin: 0 0 10px 30px;
    position: absolute;
    width: 80%;
    height: 64px;
    top: 40%;
    border: none;
}
```

```

#password-container {
  margin: 0 0 10px 30px;
  position: absolute;
  width: 80%;
  height: 64px;
  top: 60%;
  border: none;
}

input {
  width: 90%;
  padding: 10px;
  border: none;
  border-bottom: 1px solid black;
}

button {
  /* remove default behaviour */
  appearance: none;
  -webkit-appearance: none

  /* usual styles */
  padding: 20px;
  border: none;
  background-color: #3F51B5;
  color: #fff;
  font-weight: 350;
  border-radius: 1px;
  width: 20%;
  height: 30px;
  position: absolute;
  top: 80%;
  left: 65%;
}

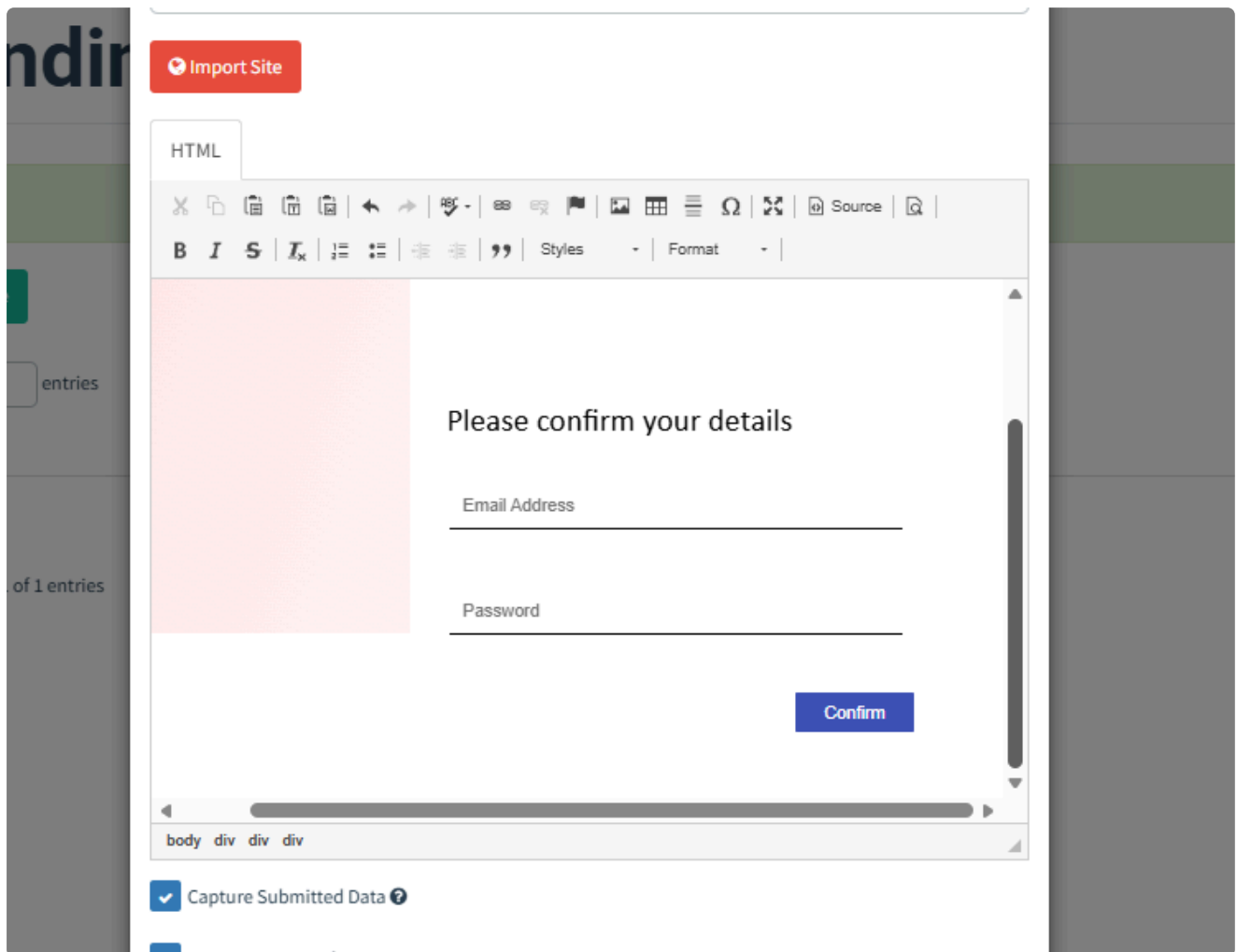
</style>
</head>
<body>
  <div id="background">
    <div id="window">
      <div id="microsoft-logo">
        <img src= /> /*Your image needs to be in Base64 format. I use
codebeautify to convert my images to Base64 format*/
      </div>
      <div id="confirm-details">
        Please confirm your details
      </div>
      <form action="text" method="POST" target="_blank">
        <div id="email-container">

```

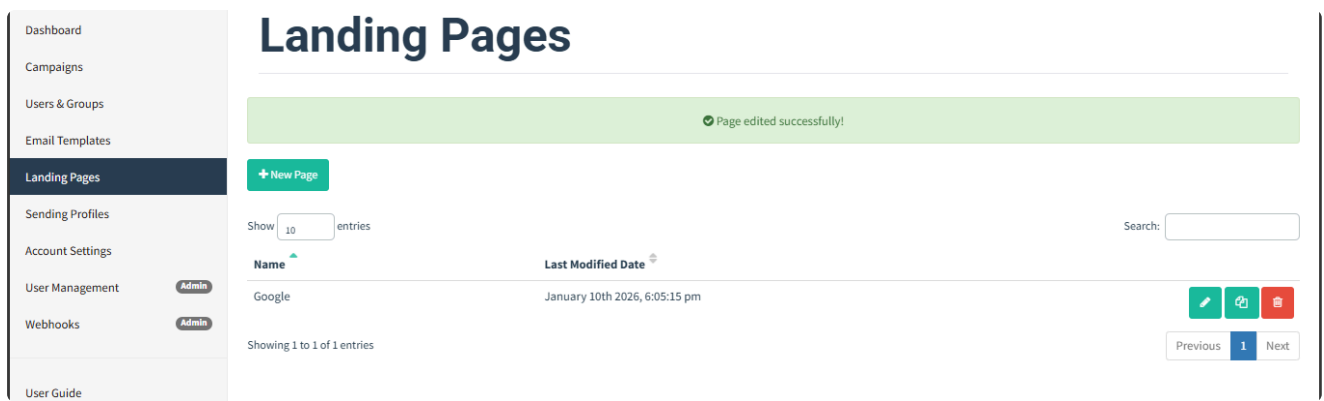
```

        <input type="text" name="email" placeholder="Email Address"/>
    </div>
    <div id="password-container">
        <input type="text" name="password" placeholder="Password"/>
    </div>
    <button onClick="closeWin()">Confirm</button>
</form>
</div>
</div>
<script>
var myWindow
function closeWin() {
    myWindow.close();
}
</script>
</body>
</html>

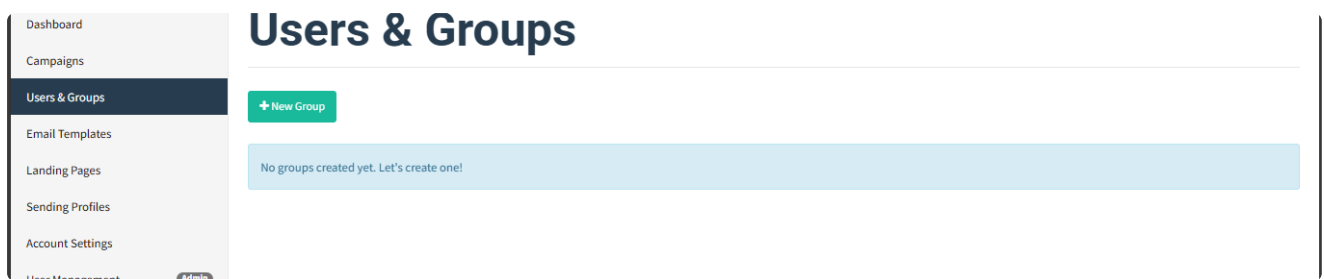
```



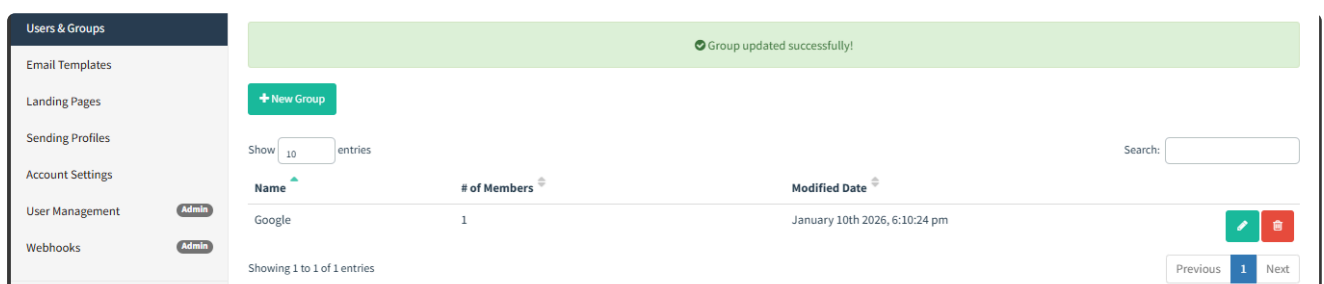
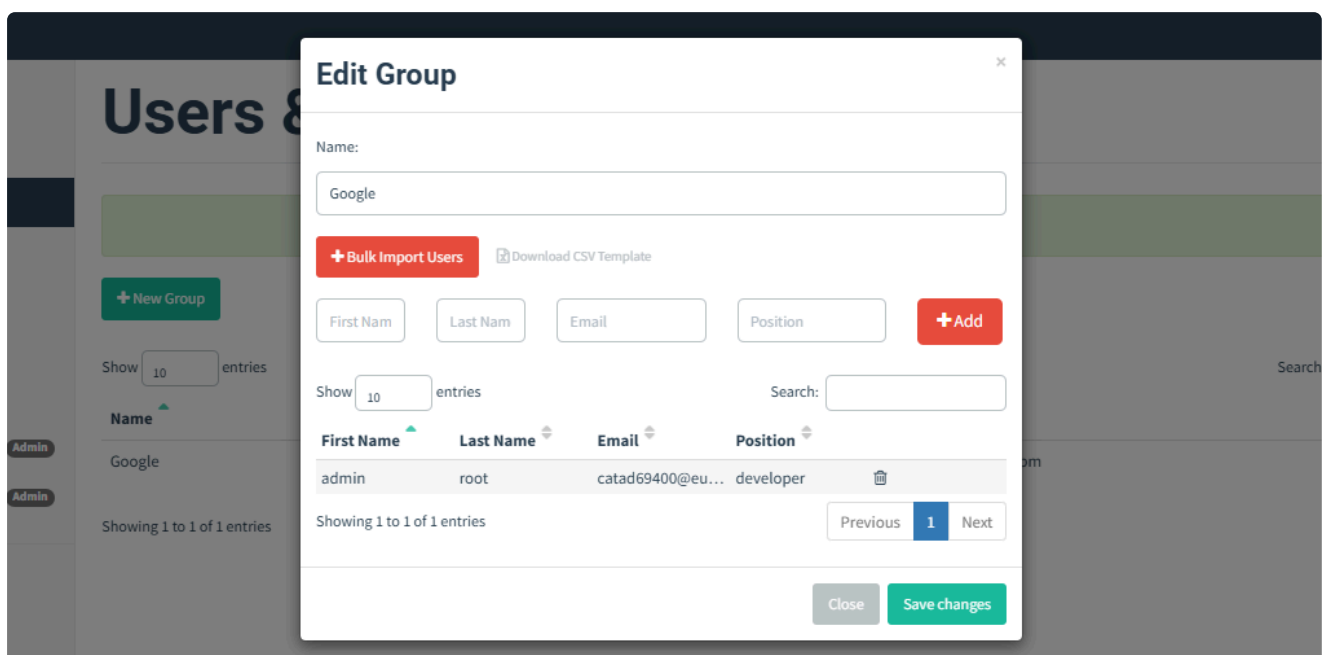
Save Page



Users & Groups : In GoPhish, the **Users & Groups** section is where you define and organize the people who will receive your phishing emails.

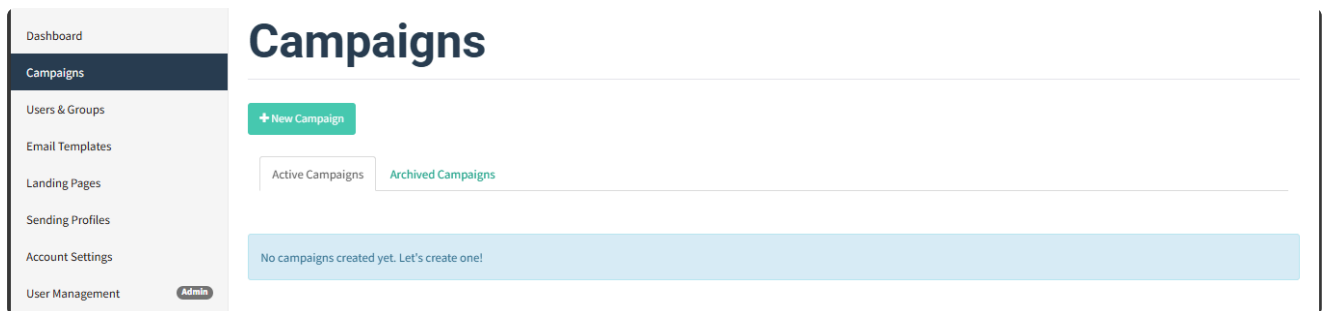


Add Details and save changes



Launch Campaign

After configuring all required components including users and groups, sending profile, email template, and landing page, the phishing campaign was launched using the Gophish dashboard. During the launch process, the campaign name, target group, email template, landing page, and sending profile were selected and reviewed for accuracy. Once launched, Gophish automatically sent phishing emails to the target users and began tracking user interactions such as email opens, link clicks, and form submissions in real time. The campaign was allowed to run for a predefined duration to collect sufficient data for analysis and reporting.



Click on New Campaign

New Campaign

Name:

admin

Email Template:

Google

Landing Page:

Google

URL: ?

http://192.168.0.100

Launch Date

January 10th 2026, 6:15 pm

Send Emails By (Optional) ?

Sending Profile:

IT Admin

Send Test Email

Groups:

Select Group

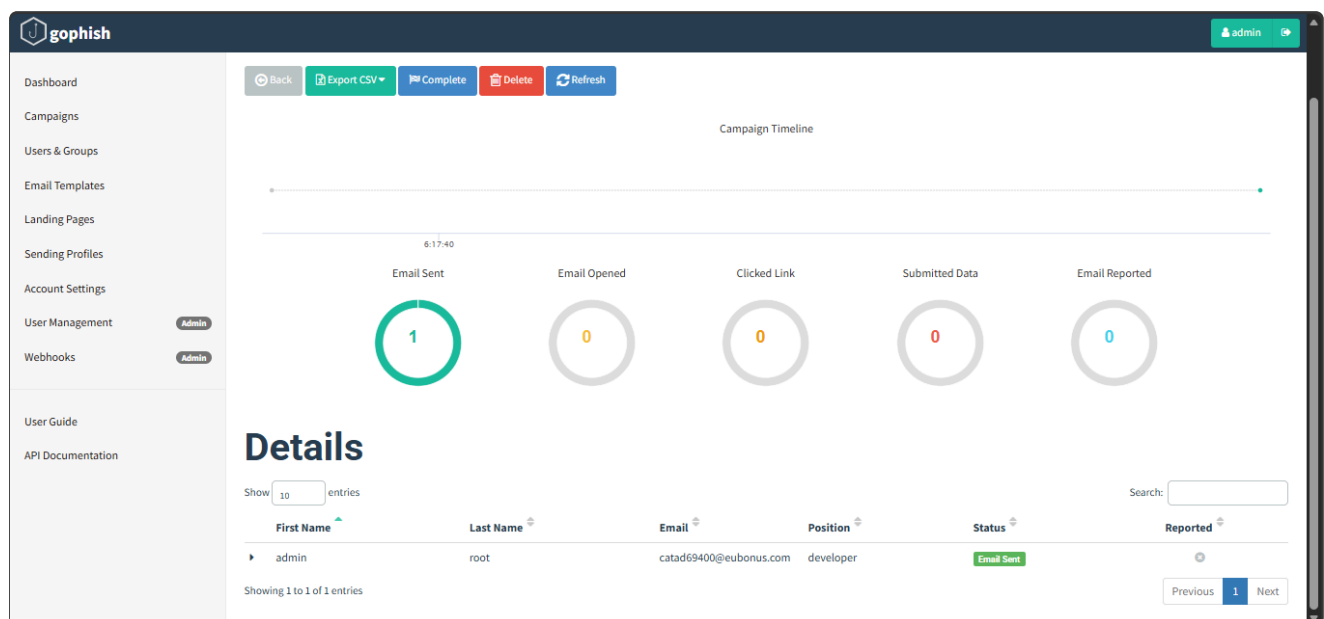
Close

Launch Campaign

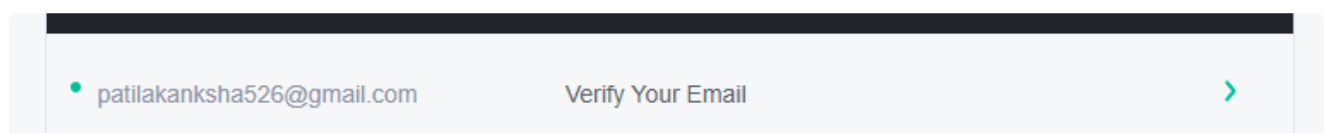
- **Campaign Name** – Used to identify and manage the phishing campaign within the Gophish dashboard.
- **Email Template** – Defines the phishing email content sent to target users.
- **Landing Page** – Specifies the web page users are redirected to after clicking the phishing link.
- **URL** – The phishing server address embedded in the email links.
- **Launch Date** – Determines when the campaign starts sending emails.
- **Sending Profile** – Configures the SMTP settings used to deliver emails.
- **Target Groups** – Contains the list of users who will receive the phishing emails.

Campaign Monitoring and Results

After launching the phishing campaign, Gophish continuously monitored user interactions in real time. The campaign dashboard displayed detailed metrics such as email delivery status, email opens, link clicks, and credential submissions. Each user action was recorded in a timeline view, allowing clear visibility into how users responded to the phishing email. These results were used to evaluate user awareness, identify high-risk users, and measure the overall effectiveness of the phishing simulation. The collected data was later used only for analysis and security awareness improvement.



After Launching Campaign Targets will get Mails related to phishing like this





patilakanksha526@gmail.com

Date:

10-01-2026 18:28:18

Subject: Verify Your Email

Dear User,
We detected unusual activity on your email account.
[Verify Email Account](#)
If not verified within 24 hours, access may be suspended.
Regards,
IT Support Team

Conclusion

This documentation presented the complete process of conducting a phishing simulation using the Gophish framework, including campaign setup, email templates, landing pages, sending profiles, campaign launch, and result monitoring. The simulated campaign demonstrated how user behavior can be analyzed in a controlled and ethical manner to identify awareness gaps. The outcomes of the campaign can be used to strengthen security policies, improve user training programs, and enhance overall organizational resilience against phishing attacks.